

Project Report — Sales Data Analysis (Power BI)

1. Executive Summary

This project delivers an interactive Power BI dashboard that cleans and models raw retail sales data and provides actionable business insights. Key outputs include cleaned data, robust DAX measures (KPIs), and interactive visuals for top products, countries, customers, and time-series trends. The dashboard is ready for presentation, publication to Power BI Service, and use in operational decision-making.

2. Objectives

- Clean and prepare raw sales records.
 - Build accurate measures (Total Sales, Orders, Quantity, Customers, AOV).
 - Visualize top products, top countries, top customers, and sales over time.
 - Produce an interactive, production-quality dashboard suitable for business stakeholders and portfolio display.
-

3. Source Data Overview

Primary fields used:

- InvoiceNo, InvoiceDate, StockCode, Description, Quantity, UnitPrice, CustomerID, Country.

Derived column:

- $TotalPrice = Quantity * UnitPrice$ (created in Power Query).

Data preparation confirmed proper data types (Date, Whole Number, Decimal, Text), removed invalid rows (negative quantities, zero prices where appropriate), and filtered out incomplete records.

4. ETL / Data Preparation Steps (Power Query)

1. Convert InvoiceDate to Date/Time.

2. Filter out rows where $\text{Quantity} \leq 0$ and $\text{UnitPrice} \leq 0$ (treat as returns/data errors).
3. Remove or flag rows with missing CustomerID if unusable.
4. Add computed column $\text{TotalPrice} = \text{Quantity} * \text{UnitPrice}$.
5. Apply and load to the model.

5. Data Model & DAX Measures

Table name used: **data**. Key DAX measures (copy-paste ready):

Total Sales = SUM(data[TotalPrice])

Total Orders = DISTINCTCOUNT(data[InvoiceNo])

Total Quantity = SUM(data[Quantity])

Total Customers = DISTINCTCOUNT(data[CustomerID])

AOV = DIVIDE([Total Sales], [Total Orders], 0)

Sales per Customer = DIVIDE([Total Sales], [Total Customers], 0)

Notes:

- DIVIDE(..., 0) protects against division-by-zero.
- All measures should be created as *Measures* (not calculated columns).

6. Visuals Built

- **KPI cards:** Total Sales, Total Orders, Total Quantity, Total Customers, AOV, Sales per Customer.
- **Slicer:** Country (dropdown) — drives all visuals.

- **Top Products by Sales:** horizontal bar chart (Top N).
 - **Top Countries by Sales:** horizontal bar chart (Top N).
 - **Sales Over Time:** Line chart with area fill (continuous monthly trend).
 - **Top Customers by Sales:** Bar chart (Top 10).
 - Design system applied: dark background, neon primary color palette, subtle glass cards, transparent plot areas, consistent spacing and padding.
-

7. Example Findings (from current dataset)

- **Total Sales:** \$10.64M
- **Total Orders:** 19.96K
- **Total Quantity:** 5.572M
- **Total Customers:** 4.34K
- **Top Country (example):** United Kingdom (highest revenue in sample)
- **Top Products (examples):** shown in Top Products visual (refer to bar chart)

These are sample results pulled from the dataset loaded into the PBIX file.